樂升半導體 LEVETOP

LT7689

High Performance Serial Uart TFT Graphics Controller

Introduction

LT7689 is an efficient serial Uart TFT panel controller. It combines a Cortex-M4 MCU and 2D TFT graphical display accelerators. Through Uart interface, **LT7689** can receive commands from the host MCU, and display predefined pictures and contents to the connected TFT panel. In addition to its high-performance M4 MCU, **LT7689** also provides graphics acceleration, PIP (Picture-in-Picture), geometry drawing and other functions. It can improve the efficiency while displaying pictures/contents to the TFT panel and reduce the time it takes for the host MCU to process graphical displays. **LT7689** supports an RGB interface display with a display resolution ranging from 320 x 240 (QVGA) to 1280 x 1024 (SXGA) of 16/18/24bits TFT RGB panel.

LT7689 has an embedded M4 core MCU, with a maximum CPU speed of 150MHz, and contains508Kbytes Flash, 256Kbytes SRAM, 2D graphics acceleration display, and 128Mb display memory. It provides more than 70 commands, including picture display, GIF animation display, loop chart display, power-on image display, progress bar display, text string display, QR code generation, audio playback, and combining touch screen to achieve touch function, etc. With the editing software and simulation tool provided by Levetop, users can easily get their project done with TFT display functions in no time. Along with the Uart interface for comunicating with the host MCU, **LT7689** also provides multiple groups of SCI (Uart) interfaces that can connect to peripherals such as blue tooth modules, and WiFi modules, etc. USB interfaces, SD cards, analog input AIN, PWM and INT interrupts are also available. These interfaces can also be set up as normal IO interfaces. **LT7689** also has an embedded RTC.

Benefiting from its high-capacity Flash and SRAM, **LT7689** can also be used as a host MCU with TFT controller. The hardware's graphics acceleration engine (BTE) and geometric drawing engine can even support display rotation, picture mirror shooting, picture-in-picture (PIP/child picture) and graphics mixed transparent display, as well as painting points, drawing lines, curves, ellipses, triangles, rectangles, rounded rectangles, cylinders, tables and other functions. **LT7689**'s powerful display capabilities are ideal for electronic products with TFT-LCD screens, such as smart home appliances, automotive dashboards, motorcycle panels, multi-function transaction machines, industrial control applications, electronic instruments, medical beauty equipment, testing equipment, charging equipment, hydropower meters, and smart audio player with TFT panel, etc.

System Application

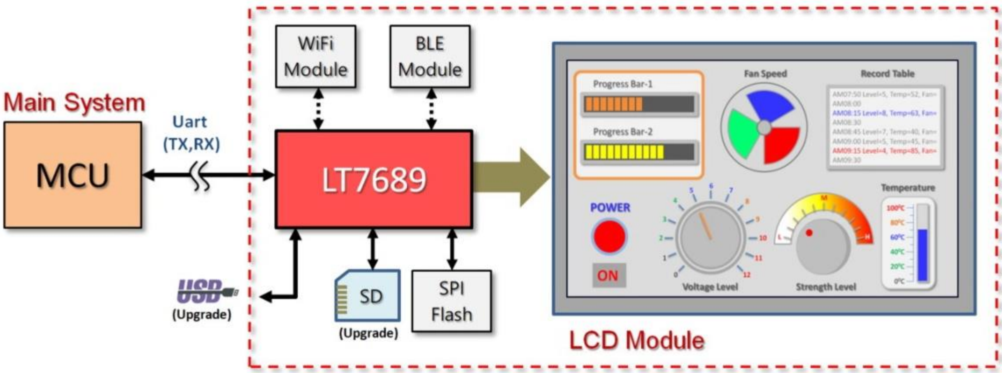

Figure 1: LT7689 Applicatio-1

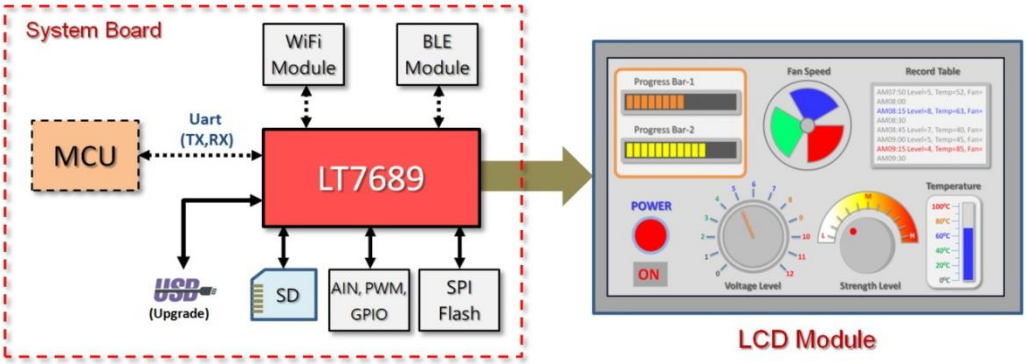

Figure 2: LT7689 Application-2

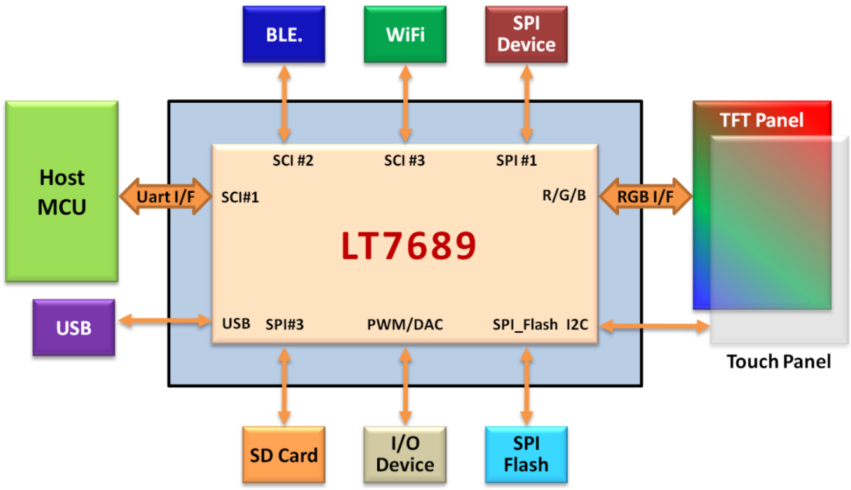

Figure 3: Applicaion Block Diagram

Internal Block

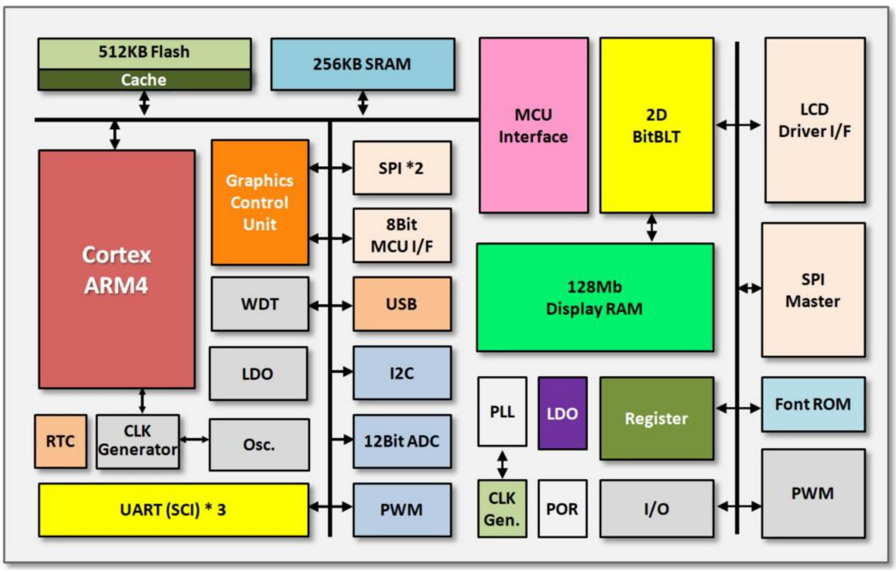

Figure 4: Internal Block Diagram

Features

System MCU Interface
- Support SCI/Uart and USB Interface.
- Embedded 32Bit Cortex-M4 with 150MHz Clock.

USB Interface
- Supports USB2.0 Full Speed.

SCI (Uart) Interface
- Support three SCI (Serial Communications Interface)

Memory
- MCU Embedded 508K bytes Flash.
- MCU Embedded 256K bytes SRAM.
- TFTController Embedded 128Mb Display RAM.

Display Data Formats
- 1bpp: Monochrome Data (1-bit/Pixel)
- 8bpp: RGB 3:3:2 (1-byte/Pixel)
- 16bpp: RGB 5:6:5 (2-byte/Pixel)
- 18bpp: RGB 8:8:8 (3bytes/Pixel).
  - Index 2:6 (64 Index Colors/Pixel with Opacity Attribute)
  - αRGB 4:4:4:4 (4096 Colors/Pixel with Opacity Attribute)

Support Panel and Resolution
- Support 16, 18, 24 bits RGB Panel.
- Supported Resolution:
  - QVGA : 320*240, 16/18-bit LCD Panel
  - WQVGA : 480*272, 16/18-bit LCD Panel
  - VGA : 640*480, 16/18-bit LCD Panel
  - WVGA : 800*480, 16/18-bit LCD Panel
  - SVGA : 800*600, 16/18-bit LCD Panel
  - WSVGA : 1024*600, 16/18-bit LCD Panel
  - XGA : 1024*768, 16/18-bit LCD Panel
  - SXGA : 1280*1024, 16/18-bit LCD Panel

Display Functions
- Multiple Display Buffer: Multi buffering allows the main display window to be switched among buffers. Multi buffering allows a simple animation display to be performed by switching the buffers
- Horizontal/Vertical Flip Display: Vertical Flip display functions are available for image data reads. PIP window will be disabled if flip display function enable
- Mirror and Rotation Functions are available for Image Data Writes
- Provide four User-defined 32*32 Pixels Graphic Cursor
- Virtual Display: Virtual display is used to show an image which is larger than LCD panel size. The image may scroll easily in any direction
- Picture-in-Picture (PIP) Display: Two PIP windows area are available: Enabled PIP windows are always displayed on top of the Main window. The PIP1 window is always on top of the PIP2 window.
- Wake-up Display: Wake-up Display is used to quickly show the display data stored in Display RAM. This feature is used when returning from Standby mode or Suspend mode.
- Color Bar: It can display color bar on panel directly. Default resolution is 640 dots by 480 dots.

Bit Block Transfer Engine (BitBLT)
- 2D BTE Engine
- Copy Image with Raster Operators
- Color Depth Conversion
- Solid Fill & Pattern Fill
- Provide User-defined Patterns with 8*8 Pixels or 16*16 Pixels
- Opacity (Alpha-Blend) Control: It blends two images and then generates a new image
  - Chroma-Keying Function: Mixes images with the specified RGB color based on the transparency rate.
  - Window Alpha-Blending Function: Mixes two images based on the transparency rate of the specified region (fade-in and fade-out functions are available).
  - Dot Alpha-Blending Function: Mixes images based on the transparency rate when the target is a graphics image in the RGB format.

Shape Drawing Engine
- Provide Smart Drawing Features: Line, Rectangle, Triangle, Polygon, Poly-Line, Circle, Ellipse, Arc, Rounded-Rectangle and Circle-Rectangle

Text Features

- Embedded 8*16, 12*24, 16*32 Character Sets of ISO/IEC 8859-1/2/4/5.
- User-defined Characters Support Half Size & Full Size for 8*16, 12*24 and 16*32.
- Programmable Text Cursor for Writing with Character.
- Character Enlargement Function *1, *2, *3, *4 for Horizontal/Vertical Direction.
- Support Character Rotates 90 Degree.

SPI Master Interface

- Provide DMA Function: Support Direct Data Transfer from External Serial Flash to Frame Buffer
- Compatible with Standard SPI Specifications
- Provide 16bytes Read FIFO and 16bytes Write FIFO
- Provide Interrupt when Tx FIFO was Empty and SPI Tx/Rx Engine Idle.
- Support Extra 2 STD SPI nterface。

I2C Interface

- MCU Provides I2C Interface.
- Support Standard Mode (100kbps) and Fast Mode (400kbps).

PWM Interface

- MCU Provides 4 PWM Output.
- TFT Controller Embedded Two 16bits Counter and 2 PWM Output.
- Programmable Duty Control of Output Waveform (PWM).

Interrupt Signals

- MCU Provides 6 Interrupt Inputs.
- TFT Controller Provides 1 Interrupt Output.

GPIO

- MCU Supports 10 GPIO.
- TFT Controller Provides 14 GPIO.

Analog Input

- MCU Provides 2 ADC's Analog Input.

Reset

- MCU Provides Power-On-Reset, External Reset, Software Reset, WatchDog Reset and Voltage Detect Reset
- TFT Controller Power-On-Reset, External Reset, Software Reset.

Power Saving Mode

- Provide Standby, Suspend and Sleep Mode.
- Support MCU Wakeup.

Clock Source

- MCU Embedded a Clock up to150MHz.
- Support 32.768Khz External Crystal Oscillator Circuit for RTC.
- Embedded Programmable PLL for Core Clock, LCD Panel's Pixel Clock and Frame Buffer Clock.

Power Supply

- VDD Power: 3.3V +/- 0.3V
- Embedded 1.1V、1.2V、1.8V LDO

Package

- QFN-96Pin（10*10mm^2）

Temperature

- -40℃~85℃。

Pin Assignment

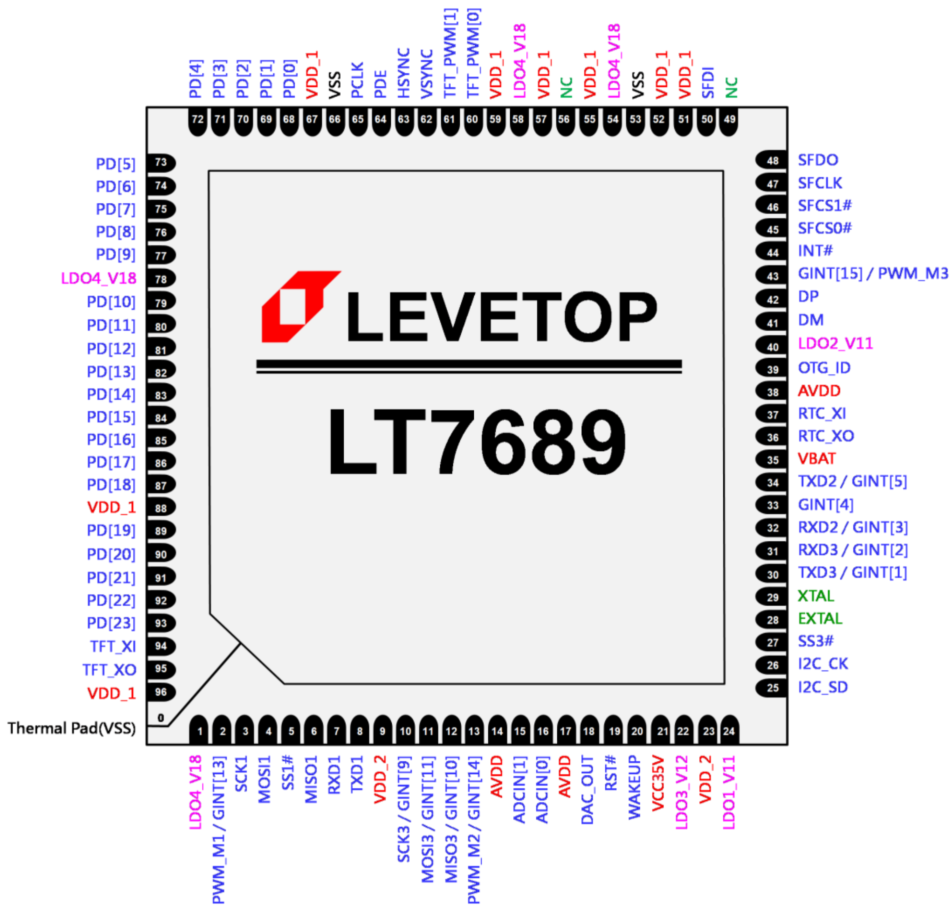

Figure 5: LT7689 Pin Assignment（QFN-96Pin)

Pin Description

SCI (Uart) Signals

Table 1: SCI (Uart) Signals

Pin #	Pin Name	I/O	Pin Description
7	RXD1	I	**Received Data Input #1** This pin is for the Received Data Input of SCI module #1. It is used to connect the external SCI device.
8	TXD1	O	**Transmittion Data Output #1** This pin is for the Transmittion Data Output of SCI module #1. It is used to connect the external SCI device.
32	RXD2	I	**Received Data Input #2** This pin is for the Received Data Input of SCI module #2. It is used to connect the external SCI device. This pin also can be used as GPIO or GINT[3].
34	TXD2	O	**Transmittion Data Output #2** This pin is for the Transmittion Data Output of SCI module #2. It is used to connect the external SCI device. This pin also can be used as GPIO or GINT[5].
31	RXD3	I	**Received Data Input #3** This pin is for the Received Data Input of SCI module #3. It is used to connect the external SCI device. This pin also can be used as GPIO or GINT[2].
30	TXD3	O	**Transmittion Data Output #3** This pin is for the Transmittion Data Output of SCI module #3. It is used to connect the external SCI device. This pin also can be used as GPIO or GINT[1].

■ LCD Driver Signals

Table 2: LCD Driver Signals

<table>
<tr><th>Pin #</th><th>Pin Name</th><th>I/O</th><th>Pin Description</th></tr>
<tr><td>93~89,
87~79,
77~68</td><td>PD[23:19],
PD[18:10],
PD[9:0]</td><td>O</td><td>LCD Panel Data Bus
TFT LCD data bus output for source driver. User can connect to corresponding RGB bus for different settings.
<table>
<tr><th rowspan="2">Pin Name</th><th colspan="3">TFT-LCD Interface</th></tr>
<tr><th>16bits</th><th>18bits</th><th>24bits</th></tr>
<tr><td>PD[0]</td><td colspan="2">GPIOD[0]</td><td>B0</td></tr>
<tr><td>PD[1]</td><td colspan="2">GPIOD[1]</td><td>B1</td></tr>
<tr><td>PD[2]</td><td>GPIOD[6]</td><td>B0</td><td>B2</td></tr>
<tr><td>PD[3]</td><td>B0</td><td>B1</td><td>B3</td></tr>
<tr><td>PD[4]</td><td>B1</td><td>B2</td><td>B4</td></tr>
<tr><td>PD[5]</td><td>B2</td><td>B3</td><td>B5</td></tr>
<tr><td>PD[6]</td><td>B3</td><td>B4</td><td>B6</td></tr>
<tr><td>PD[7]</td><td>B4</td><td>B5</td><td>B7</td></tr>
<tr><td>PD[8]</td><td colspan="2">GPIOD[2]</td><td>G0</td></tr>
<tr><td>PD[9]</td><td colspan="2">GPIOD[3]</td><td>G1</td></tr>
<tr><td>PD[10]</td><td>G0</td><td>G0</td><td>G2</td></tr>
<tr><td>PD[11]</td><td>G1</td><td>G1</td><td>G3</td></tr>
<tr><td>PD[12]</td><td>G2</td><td>G2</td><td>G4</td></tr>
<tr><td>PD[13]</td><td>G3</td><td>G3</td><td>G5</td></tr>
<tr><td>PD[14]</td><td>G4</td><td>G4</td><td>G6</td></tr>
<tr><td>PD[15]</td><td>G5</td><td>G5</td><td>G7</td></tr>
<tr><td>PD[16]</td><td colspan="2">GPIOD[4]</td><td>R0</td></tr>
<tr><td>PD[17]</td><td colspan="2">GPIOD[5]</td><td>R1</td></tr>
<tr><td>PD[18]</td><td>GPIOD[7]</td><td>R0</td><td>R2</td></tr>
<tr><td>PD[19]</td><td>R0</td><td>R1</td><td>R3</td></tr>
<tr><td>PD[20]</td><td>R1</td><td>R2</td><td>R4</td></tr>
<tr><td>PD[21]</td><td>R2</td><td>R3</td><td>R5</td></tr>
<tr><td>PD[22]</td><td>R3</td><td>R4</td><td>R6</td></tr>
<tr><td>PD[23]</td><td>R4</td><td>R5</td><td>R7</td></tr>
</table>
When the LCD is set to 16/18bpp function mode, some PDs can be defined as GPIO pins.</td></tr>
<tr><td>65</td><td>PCLK</td><td>O</td><td>Pixel Clock
Generic TFT interface signal for panel scan clock. It derives from internal PLL.</td></tr>
</table>

Pin #	Pin Name	I/O	Pin Description
62	VSYNC	O	**VSYNC Pulse** Generic TFT interface signal for vertical synchronous pulse.
63	HSYNC	O	**HSYNC Pulse** Generic TFT interface signal for horizontal synchronous pulse.
64	PDE	O	**Data Enable** Generic TFT interface signal for data valid or data enable.

SPI Signals

Table 3: SPI Signals

Pin #	Pin Name	I/O	Pin Description
3	SCK1	O	**SPI #1 Clock Signal** This is the clock output pin for the 1st SPI. It is connected to the the external SPI device.
5	SS1#	O	**SPI #1 Chip Select** This is the chip select output pin for the 1st SPI.
4	MOSI1	O	**SPI #1 Data Output** This pin is to output data signal to external SPI device. (Master Out Slave In)
6	MISO1	I	**SPI #1 Data Input** This pin is to read data input from external SPI device. (Master Int Slave Out)
10	SCK3 GINT[9]	O	**SPI #3 Clock Signal** This is the clock output pin for the 3rd SPI. It is connected to the external SPI device. This pin also can be used as GINT[9].
27	SS3#	O	**SPI #3 Chip Select** This is the chip select output pin for the 3rd SPI.
11	MOSI3 GINT[11]	O	**SPI #3 Data Output** This pin is to output data signal to external SPI device. (Master Out Slave In) This pin also can be used as GINT[11].
12	MISO3 GINT[10]	I	**SPI #3 Data Input** This pin is to read data input from external SPI device. (Master Int Slave Out) This pin also can be used as GINT[10].

Serial SPI Flash Signals

Table 4: Serial SPI Flash Signals

Pin #	Pin Name	I/O	Pin Description
45	SFCS[0]# GPIOC[3]	IO	**Chip Select 0 for External Serial Flash or SPI device** SPI Chip select pin #0 for serial Flash or SPI device. If SPI master I/F is disabled then it can be programmed as GPIOC[3], and default is input function.
46	SFCS[1]# GPIOC[4]	IO	**Chip Select 1 for External Serial Flash or SPI device** SPI Chip select pin #1 for serial Flash or SPI device. If SPI master I/F is disabled then it can be programmed as GPIOC[4], and default is input function.
47	SFCLK GPIOC[0]	IO	**SPI Serial Clock** Serial clock output for serial Flash/ROM or SPI device. If SPI master I/F is disabled then it can be programmed as GPIO C[0], and default is input function.
48	SFDO GPIOC[1]	IO	**Master Output Slave Input** **Single Mode**: Data input of serial Flash or SPI device. For LT7689, it is output. **Dual Mode**: The signal is used as bi-direction data #0(SIO0). Only valid in serial flash DMA mode. If SPI master I/F is disabled then it can be programmed as GPIO C[1], and default is input function.
50	SFDI GPIOC[2]	IO	**Master Input Slave Output** **Single Mode**: Data output of serial Flash or SPI device. For LT7689, it is input. **Dual Mode**: The signal is used as bi-direction data #1(SIO1). Only valid in serial flash DMA mode. If SPI master I/F is disabled then it can be programmed as GPIOC[2], and default is input function.

PWM Signals

Table 5: PWM Signals

Pin #	Pin Name	I/O	Pin Description
60	TFT_PWM[0] GPIOC[7]	IO	**TFTC PWM #0 Output** This signal is controlled by the registers of LT7689's internal TFT controller. It is a programmable PWM output signal that can be used to control the backlight of the TFT panel or other devices. The output mode of the TFT_PWM[0] can be set via the registers of the TFT controller. This pin also can be used as GPIOC[7].
61	TFT_PWM[1]	IO	**TFTC PWM #1 Output** This signal is controlled by the registers of LT7689's internal TFT controller. It is a programmable PWM output signal that can be used to control the backlight of the TFT panel or other devices. The output mode of the TFT_PWM[1] can be set via the registers of the TFT controller.
2	PWM_M1 GINT[13]	IO	**MCU-controlled PWM Output Signal** It can be used as a PWM output or GPIO and is set by internal MCU registers. Normally, this pin outputs a fixed clock signal (8~12MHz) to the TFT clock signal input port – TFT_XI. This pin also can be used as GINT[13].
13	PWM_M2 GINT[14]	IO	**MCU-controlled PWM Output Signal** It can be used as a PWM output or GPIO and is set by internal MCU registers. This pin also can be used as GINT[14].
43	PWM_M3 GINT[15]	IO	**MCU-controlled PWM Output Signal** It can be used as a PWM output or GPIO and is set by internal MCU registers. This pin also can be used as GINT[15].

USB Signals

Table 6: USB Signals

Pin #	Pin Name	I/O	Pin Description
42	DP	IO	**USB Data Positive** These signals are used by the USB module.
41	DM	IO	**USB Data Negative** These signals are used by the USB module.
39	OTG_ID	IO	**USB Detect** This ID signal is used to indicate the ID status of the Mini USB plug, and to judge if it is a master device or a slave device.

GPIO and Interrupt Signals

Table 7: GPIO and Interrupt Signals

Pin #	Pin Name	I/O	Pin Description
30	GINT[1]	IO	**GPIO/INT #1 Signal** This pin is used as GPIO or interrupt input signal GINT[1]. It also can be used as TXD3.
31	GINT[2]	IO	**GPIO/INT #2 Signal** This pin is used as GPIO or interrupt input signal GINT[0]. It also can be used as TRD3.
32	GINT[3]	IO	**GPIO/INT #3 Signal** This pin is used as GPIO or interrupt input signal GINT[0]. It also can be used as RXD2.
33	GINT[4]	IO	**GPIO/INT #4 Signal** This pin is used as GPIO or interrupt input signal GINT[4].
35	GINT[5]	IO	**GPIO/INT #5 Signal** This pin is used as GPIO or interrupt input signal GINT[5]. It also can be used as TXD2.。
10	GINT[9]	O	**GPIO/INT #9 Signal** This pin is used as GPIO or interrupt input signal GINT[9]. It also can be used as SCK3.
11	GINT[11]	O	**GPIO/INT #11 Signal** This pin is used as GPIO or interrupt input signal GINT[11]. It also can be used as MOSI3.
12	GINT[10]	I	**GPIO/INT #10 Signal** This pin is used as GPIO or interrupt input signal GINT[10]. It also can be used as MISO3.
2	GINT[13]	IO	**GPIO/INT #13 Signal** This pin is used as GPIO or interrupt input signal GINT[13]. It also can be used as PWM_M1.
13	GINT[14]	IO	**GPIO/INT #14 Signal** This pin is used as GPIO or interrupt input signal GINT[14]. It also can be used as PWM_M2.
43	GINT[15]	IO	**GPIO/INT #15 Signal** This pin is used as GPIO or interrupt input signal GINT[15]. It also can be used as PWM_M3.
44	INT#	O	**Interrupt Output Signal of TFTC** This pin is to indictate the interrupt status of the TFT controller. If an interrupt occurred, then this pin will be pulled low.
60, 46, 45, 50, 48, 47	GPIOC[7] GPIOC[4:0]	IO	**GPIO C Group** These are general purpose I/O. GPIOC are available when PWM and SPI Master functions disabled. GPIOC[7] is the same pin with PWM[0]. GPIOC[4:0] are multiplex pins that share with {SFCS1#, SFCS0#, SFDI, SFDO, SFCLK}

Pin #	Pin Name	I/O	Pin Description
87, 70	GPIOD[7:6]	IO	**GPIO D Group** These are general purpose I/O. GPIOD[7] is a multiplex pin that shares with PD[18], and GPIOD[6] is a multiplex pins that shares with PD[2]. GPIOD[7,6] are available when LCD Panel interface is set to 16bits.

Miscellaneous Signals

Table 8: Miscellaneous

Pin #	Pin Name	I/O	Pin Description
26	I2C_CK	IO	**I2C Clock** This signal is the I2C clock signal, and normally used for capacitive touch panel.
25	I2C_SD	IO	**I2C Data** This signal is the I2C data signal, and normally used for capacitive touch panel.
18	DAC_OUT	O	**DAC (Digital-to-Analog Converter) Output** This pin is the analog output of DAC, and controlled by internal MCU register. Normally used for audio output.
16 15	ADCIN[0] ADCIN[1]	I	**ADC (Analog-to-Digital Converter) Input** These pins are the analog input of ADC, and controlled by internal MCU register.
20	WAKEUP	I	**Wake-up Input Signal** This is LT7689 wake-up input signal **SD card (SPI mode) detect signal** When SD card (SPI mode) is used, this signal can be used as the SD card (SPI mode) detection signal.
19	RST#	I	**Reset** When RST# = 0, the internal MCU will be reset.

Power and Clock Signals

Table 9: Power and Clock Signals

Pin #	Pin Name	I/O	Pin Description
94	TFT_XI	I	**TFTC Crystal / External Clock Input** This pin is used to connect to a clock souce for the internal TFTC's PLL. If an external crystal or clock source is used, it should be connected to this pin. In general, this pin is connected to PWM_M1, and the susgested frequency is 8~12MHz.
95	TFT_XO	O	**TFTC Crystal Output** This is an output pin for internal crystal circuit of TFT Controller. It should be connected to an external crystal circuit.
37	32K_XI	I	**32.768Khz Crystal Input** This pin is connected to 32.768Khz X'tal.
36	32K_XO	O	**32.768Khz Crystal Output** This pin is connected to 32.768Khz X'tal.
29	XTAL	I	**12Mhz Crystal Input** This pin is connected to 12Mhz X'tal for USB.
28	EXTAL	O	**12Mhz Crystal Output** This pin is connected to12Mhz X'tal for USB.
35	VBAT	PWR	**3.3V~3.6V RTC Battery Power**
21	VCC35	PWR	**3~5V Power Supply Input**
51, 52, 55, 57, 59, 67, 88, 96	VDD_1	PWR	**3.3V Power Input for TFT Controller**
9, 23	VDD_2	PWR	**3.3V Power Input for MCU**
14, 17, 38	AVDD		**3.3V Power Input for Internal Analog Circuit**
24	LDO1_V11	PWR	**Internal LDO Output #1 (1.1V, for Flash)** These pins must connect 1uF and 0.1uF capacitor to ground.
40	LDO2_V11	PWR	**Internal LDO Output #2 (1.1V for USB)** These pins must connect 1uF and 0.1uF capacitor to ground.
22	LDO3_V12	PWR	**Internal LDO Output #3 (1.2V, for Core)** These pins must connect 1uF and 0.1uF capacitor to ground.
1, 54, 58, 78,	LDO4_V18	PWR	**Internal LDO Output #3 (1.8V, for TFTC)** These pins must connect 1uF and 0.1uF capacitor to ground.
53, 66	VSS	PWR	**Ground Pins**
0	Thermal Pad	-	The back of LT7689 Heat Sink Pad must be grounded.

Function Description

LT7689 combines high-performance Cortex-M4 MCU and TFT graphics accelerators. Its internal MCU's main structure is the same as Levetop's LT32U03, and its functions can be referenced to LT32U03 specification and application manual. In addition, the TFT Graphics Accelerator applies Levetop's LT768 hardware architecture, its circuit configuration to the MCU is shown in the following figure.

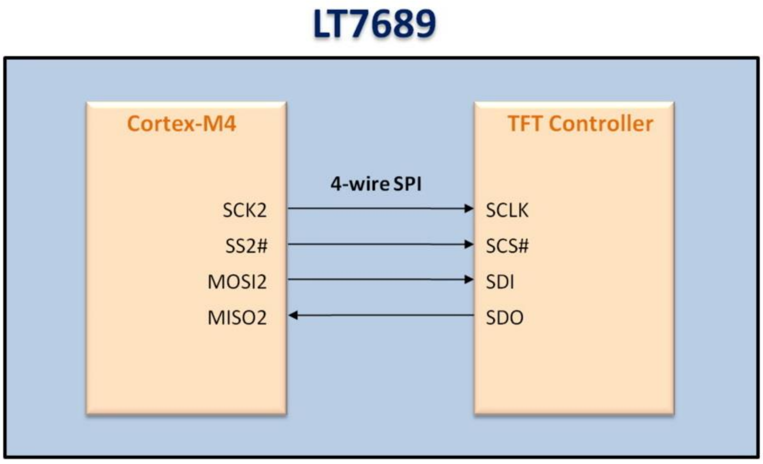

Figure 6: The connection of Internal MCU and TFT Controller

LT7689 internal MCU program already contains a Uart protocol that supports about 70 Levetop's Serial TFT panel instructions. Users can design their display flow and contents simply by using the development software provided by Levetop Semiconductor to gather pictures, animation, and other display contents together. Basically, there is no need to modify **LT7689** internal MCU program, nor need to understand **LT7689** internal register and control method. The host MCU only needs to send instructions to **LT7689** through Uart Interface, and to receive and interpret the feedback message sent by **LT7689**. Therefore, using **LT689** can save users a lot of time on display develompment tasks. For more details about development software and external application schematic, please refer to the LT7689 Uart TFT panel application note - "LT7689_UartTFT_AP Note_Vxx_EN.pdf".

Figure 7: Serial Usrt TFT Module

■ Host MCU and TFT Protocol List

Table 10: Protocol List

Main Function	Detail Function	Host Transfer (Panel Receive)						Host Receive (Panel Send)					
		Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	parameter	CRC Code (2Bytes)	End Code (4Bytes)	Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	Parameter (1Byte)	CRC Code (2Bytes)	End Code (4Bytes)
Picture Display	Display Pictures	Start	80h	nn		CRC	End	Start	80h	nn	info code	CRC	End
	Cancel Display	Start	85h	nn		CRC	End	Start	84h	nn	info code	CRC	End
	Single/Multi Picture	Start	8Ah	nn		CRC	End	Start	8Ah	nn	info code	CRC	End
	Display Single Picture	Start	8Fh	nn	X, Y, PNG, Pnn	CRC	End	Start	8Fh	nn	info code	CRC	End
	Play In Loop	Start	81h	nn		CRC	End	Start	81h	nn	info code	CRC	End
	Cancel Play In Loop	Start	84h	nn		CRC	End	Start	84h	nn	info code	CRC	End
	Transparent Pictures	Start	82h	nn		CRC	End	Start	82h	nn	info code	CRC	End
	Display GIF	Start	88h	nn		CRC	End	Start	88h	nn	info code	CRC	End
	Cancel GIF	Start	89h	nn		CRC	End	Start	89h	nn	info code	CRC	End
	Buffer Area	Start	8Eh		0, 1	CRC	End	Start	8Eh	00	info code	CRC	End
	Pop Up Picture	Start	D8h	nn		CRC	End	Start	D8h	nn	info code	CRC	End
	Scroll Picture	Start	D9h	nn		CRC	End	Start	D9h	nn	info code	CRC	End
	Cancel Scroll In Loop	Start	DBh	nn		CRC	End	Start	DBh	nn	info code	CRC	End
	Numeric Picture	Start	90h	nn	ddd.d	CRC	End	Start	90h	nn	info code	CRC	End
	True Color Numeric Picture	Start	91h	nn	ddd.d	CRC	End	Start	91h	nn	info code	CRC	End
	Ascii Picture	Start	9Fh	nn	ddd.d	CRC	End	Start	9Fh	nn	信息码	CRC	End
Display Control Button	Full Screen Slide Picture	Start	B4h	nn		CRC	End	Start	B4h	nn	info code	CRC	End
	Display Single Control Button	Start	A0h	nn		CRC	End	Start	A0h	nn	info code	CRC	End
		When Press Control Button						Start	A0h	nn	31h	CRC	End
		When Release Control Button						Start	A0h	nn	30h	CRC	End
	Cancel Display	Start	A1h	Nn		CRC	End	Start	A1h	nn	info code	CRC	End
	Virtual Control Display	Start	A2h	Nn		CRC	End	Start	A2h	nn	info code	CRC	End
		When Press Virtual Button						Start	A2h	nn	31h	CRC	End
		When Release Virtual Button						Start	A2h	nn	30h	CRC	End
	Cancel Display	Start	A3h	nn		CRC	End	Start	A3h	nn	info code	CRC	End
	Display Base Map &All Control Button	Start	9Ch	00		CRC	End	Start	9Ch	00	info code	CRC	End
		After Slide Screen						Start	9Ch	Page No.	info code	CRC	Start
		When Press Control Button						Start	9Bh	Icon ID	31h	CRC	End
		When Release Control Button						Start	9Bh	Icon ID	30h	CRC	End

Main Function	Detail Function	Host Transfer (Panel Receive)						Host Receive (Panel Send)					
		Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	parameter	CRC Code (2Bytes)	End Code (4Bytes)	Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	Parameter (1Byte)	CRC Code (2Bytes)	End Code (4Bytes)
Index Picture	Progress Bar	Start	B0h	nn	Value (2 Bytes)	CRC	End	Start	B0h	nn	info code	CRC	End
	Pointer	Start	B1h	nn	Angle (2 Bytes)	CRC	End	Start	B1h	nn	info code	CRC	End
	Ring Pointer	Start	DCh	nn	S_Angle, A_Angle	CRC	End	Start	DCh	nn	info code	CRC	End
	QR Code	Start	98h	nn	String	CRC	End	Start	98h	nn	info code	CRC	End
	Curve Buffer	Start	B2h	Chanel (0~3)	Data (n*2bytes)	CRC	End	Start	B2h	Chanel (0~3)	信息码	CRC	End
	Display Curve	Start	B3h	nn		CRC	End	Start	B3h	nn	信息码	CRC	End
Touch Slider	Set Slider	Start	94h	nn		CRC	End	Start	94h	nn	info code	CRC	End
		When Press Slider						Start	94h	nn	Value (1 Byte)	CRC	End
	Remove Slider	Start	95h	Nn		CRC	End	Start	95h	nn	info code	CRC	End
	Set Ring Slider	Start	96h	Nn		CRC	End	Start	96h	nn	info code	CRC	End
		When Press Ring Slider						Start	96h	nn	Value (1 Byte)	CRC	End
	Remove Ring Slider	Start	97h	nn		CRC	End	Start	97h	nn	info code	CRC	End
Font	Font-1	Start	C0h	nn	String	CRC	End	Start	C0h	nn	info code	CRC	End
	Font-2	Start	C1h	nn	String	CRC	End	Start	C1h	nn	info code	CRC	End
	Font-3	Start	C2h	nn	String	CRC	End	Start	C2h	nn	info code	CRC	End
	Font-4	Start	C3h	nn	String	CRC	End	Start	C3h	nn	info code	CRC	End
	Bold Font-1	Start	D0h	nn	String	CRC	End	Start	D0h	nn	info code	CRC	End
	Bold Font-2	Start	D1h	nn	String	CRC	End	Start	D1h	nn	info code	CRC	End
	Bold Font-3	Start	D2h	nn	String	CRC	End	Start	D2h	nn	info code	CRC	End
	Bold Font-4	Start	D3h	nn	String	CRC	End	Start	D3h	nn	info code	CRC	End
Cursor	Cursor On/Off	Start	86h		00/01/02	CRC	End	Start	86h	nn	info code	CRC	End
	Display Cursor	Start	87h	N	X, Y	CRC	End	Start	87h	N	info code	CRC	End
Backlight Brightness	Set Brightness	Start	BAh		BL (00~0Fh)	CRC	End	Start	BAh	BL (00~0Fh)	info code	CRC	End
	On/Off	Start	BCh		00 or 01	CRC	End	Start	BCh	00 or 01	info code	CRC	End
Wav	Play	Start	B8h		REP(Bit7) + WAV No.	CRC	End	Start	B8h	REP(Bit7) + WAV No	info code	CRC	End
	Stop	Start	B9h			CRC	End	Start	B9h	00	info code	CRC	End
Boot	Boot	Start	9Ah	00		CRC	End	Start	9Ah	00	info code	CRC	End
Combine	Command Combine	Start	9Ah	nn		CRC	End	Start	9Ah	nn	info code	CRC	End

Main Function	Detail Function	Host Transfer (Panel Receive)						Host Receive (Panel Send)					
		Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	parameter	CRC Code (2Bytes)	End Code (4Bytes)	Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	Parameter (1Byte)	CRC Code (2Bytes)	End Code (4Bytes)
Verify	TFT Verify	Start	8Bh			CRC	End	Start	8Bh	00	info code	CRC	End
Tft Reset	Reset LT7689	Start	BDh			CRC	End	Start	BDh	00	info code	CRC	End
Geometric	Dot	Start	DFh	nn	X,Y	CRC	End	Start	DFh	nn	info code	CRC	End
	Line	Start	E0h	nn		CRC	End	Start	E0h	nn	info code	CRC	End
	Hollow Circle	Start	E1h	nn		CRC	End	Start	E1h	nn	info code	CRC	End
	Solid Circle	Start	E2h	nn		CRC	End	Start	E2h	nn	info code	CRC	End
	Solid Circle With Frame	Start	E3h	nn		CRC	End	Start	E3h	nn	info code	CRC	End
	Hollow Ellipse	Start	E4h	nn		CRC	End	Start	E4h	nn	info code	CRC	End
	Solid Ellipse	Start	E5h	nn		CRC	End	Start	E5h	nn	info code	CRC	End
	Solid Ellipse With Frame	Start	E6h	nn		CRC	End	Start	E6h	nn	info code	CRC	End
	Hollow Rectangle	Start	E7h	nn		CRC	End	Start	E7h	nn	info code	CRC	End
	Solid Rectangle	Start	E8h	nn		CRC	End	Start	E8h	nn	info code	CRC	End
	Rectangle With Frame	Start	E9h	nn		CRC	End	Start	E9h	nn	info code	CRC	End
	Hollow Rounded Rectangle	Start	EAh	nn		CRC	End	Start	EAh	nn	info code	CRC	End
	Solid Rounded Rectangle	Start	EBh	nn		CRC	End	Start	EBh	nn	info code	CRC	End
	Rounded Rectangle With Frame	Start	ECh	nn		CRC	End	Start	ECh	nn	info code	CRC	End
	Hollow Triangle	Start	EDh	nn		CRC	End	Start	EDh	nn	info code	CRC	End
	Solid Triangle	Start	EEh	nn		CRC	End	Start	EEh	nn	info code	CRC	End
	Triangle With Frame	Start	EFh	nn		CRC	End	Start	EFh	nn	info code	CRC	End
	Hollow Quadrangle	Start	F0h	nn		CRC	End	Start	F0h	nn	info code	CRC	End
	Solid Quadrangle	Start	F1h	nn		CRC	End	Start	F1h	nn	info code	CRC	End
	Hollow Pentagon	Start	F2h	nn		CRC	End	Start	F2h	nn	info code	CRC	End
	Solid Pentagon	Start	F3h	nn		CRC	End	Start	F3h	nn	info code	CRC	End
	Cylinder	Start	F4h	nn		CRC	End	Start	F4h	nn	info code	CRC	End
	Square Cylinder	Start	F5h	nn		CRC	End	Start	F5h	nn	info code	CRC	End
	Table Window	Start	F6h	nn		CRC	End	Start	F6h	nn	info code	CRC	End
Numeric Keyboard	Display Numeric Keyboard	Start	A4h	00		CRC	End	Start	A4h	nn	info code	CRC	End
		When Press Keyboard						Start	A4h	nn	ASCII + info code	CRC	End
		When Press CR Key						Start	A4h	nn	ASCII + info code+ content	CRC	End

Main Function	Detail Function	Host Transfer (Panel Receive)						Host Receive (Panel Send)					
		Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	parameter	CRC Code (2Bytes)	End Code (4Bytes)	Initial Code (1Byte)	Comm. Code (1Byte)	No. (1Byte)	Parameter (1Byte)	CRC Code (2Bytes)	End Code (4Bytes)
	Cancel Keyboard	Start	A5h	00		CRC	End	Start	A5h	nn	info code	CRC	End
Full Keyboard	Display Full Keyboard	Start	A6h	00		CRC	End	Start	A6h	nn	info code	CRC	End
		When press CR key						Start	A6h	nn	info code+ (ASCII+GB2312)	CRC	End
	Cancel Full Keyboard	Start	A7h	00		CRC	End	Start	A7h	nn	info code	CRC	End
TFT Detect	Online Detect	Start	BEh			CRC	End	Start	BEh	00	5Ah, or 55h	CRC	End
	Version Detect	Start	BFh			CRC	End	Start	BFh	MCU Code(5) + Module Info. (42)	info code	CRC	End
Clock Setting	Set Clock	Start	8Ch		Y, M, D, H, M, S, W (7 Bytes)	CRC	End	Start	8Ch	00	info code	CRC	End
	Read Clock	Start	8Dh			CRC	End	Start	8Dh	Y, M, D, H, M, S, W (8)	info code	CRC	End
Display Clock	Display Digital Time, Date	Start	92h	nn		CRC	End	Start	92h	nn	info code	CRC	End
	Display Analog Time, Date	Start	93h	nn		CRC	End	Start	93h	nn	info code	CRC	End
	Display Week	Start	9Dh	nn		CRC	End	Start	9Dh	nn	info code	CRC	End
Register	Execute 9A Command	Start	CAh	Reg		CRC	End	Start	CAh	nn	info code	CRC	End
	Set Register	Start	CBh	Reg		CRC	End	Start	CBh	nn	info code	CRC	End
	Write Data	Start	CCh	Data		CRC	End	Start	CCh	nn	info code	CRC	End
	Read Data	Start	CDh	00		CRC	End	Start	CDh	Data	info code	CRC	End
	Register Data Plus 1	Start	CEh	Reg		CRC	End	Start	CEh	nn	info code	CRC	End
	Register Data Minus 1	Start	CFh	Reg		CRC	End	Start	CFh	nn	info code	CRC	End

RS-232(UART) Communication Protocol

When the host MCU sends commands to LT7689 through the UART port, in addition to the Command Code, Serial Number, and Command Parameters, it should also add 1 Byte of Start Code (fixed as 0xAA), 2 Bytes of CRC Code and 4 Bytes of End Code (fixed as 0xE4, 0x1B, 0x11 and 0xEE). The command information is shown in Table 11:

Table 11: Receive Command Information

Start Code	Command Code	Serial Number	Command Parameters	CRC	End Code
0xAA (1 Byte)	1 Byte	1 Byte	n Bytes	2 Bytes	**0xE4, 0x1B, 0x11, 0xEE** (4 Bytes)

LT7689 will respond 10 Bytes messages after receiving commands from host MCU, including Start Code, Command Code, Serial Number, Information Code, CRC Code, and End Code. The first Byte is the Start Code, followed by the command received, the third is Serial Number. Then the forth is Information Code that returns the execution result of the panel, the fifth and sixth are CRC codes, and the last four Bytes are the End Codes.

Table 12: Feedback Information

Start Code	Command Code	Serial Number	Information Code	CRC Code	End Code
0xAA (1 Byte)	1 Byte	Normal command (1 Byte) 8Dh command (8 Bytes) BFh command (47 Bytes)	1Byte 0x00: Command executed 0x01: Command parameter error 0x02: Non-exist command 0x03: Command's configuration Overflow 0x04: CRC correction error 0x05: Flash Data Exception BEh command: 0x5A: Ready 0x55: Not Ready 94h 94h Command: the percentage position of whole progress bar A0h, 9Ch/9Bh Command: 0x31: Press Icon 0x30: Release Icon	2 Bytes	0xE4, 0x1B, 0x11, 0xEE (4 Bytes)

In the feedback information, serial numbers may represent different meanings in some commands. For example,in 9Ch command, it means the page number; in BAh command, it means the backlight brightness; in B8h, it means WAV No., in 8Dh, it has 8 Bytes and means the clock information; in BFh command, it has 47 Bytes which represent the TFT panel information. Please refer to **LT7689** Application Note for detail.

Package Information

■ LT7689 (QFN-96pin)

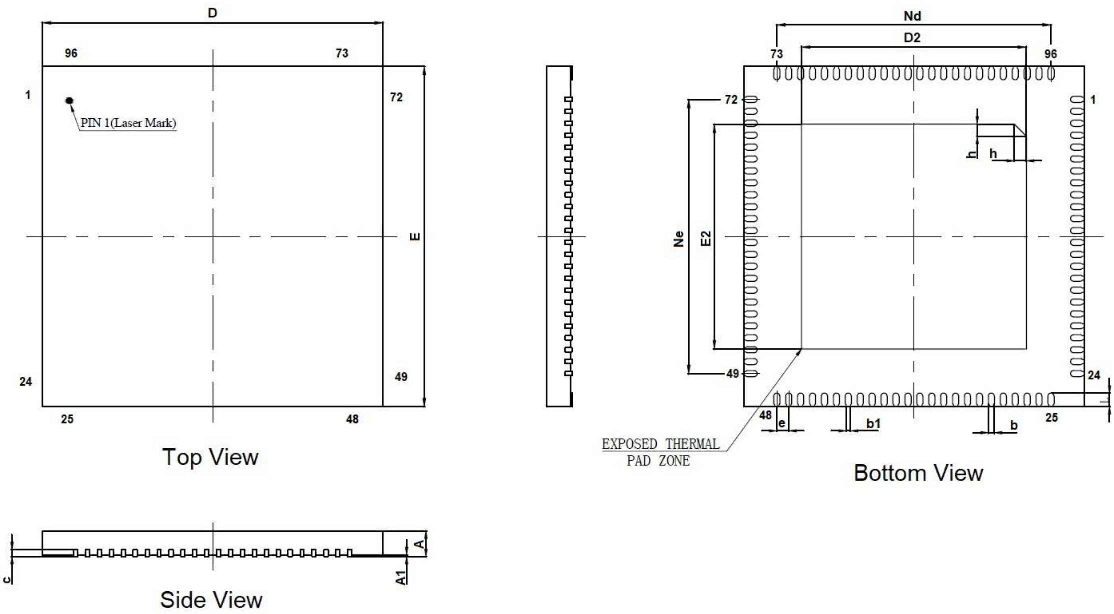

Figure 8: QFN-96Pin Outline

Note: When designing the PCB layout, the heat pad of LT7689's back (thermal Pad Zone) must be directly grounded.

Table 13: QFN-96Pin Dimenstion

Symbol	Millimeter			Symbol	Millimeter		
	Min.	Nom.	Max		Min.	Nom.	Max
A	0.80	0.85	0.9	E	9.90	10.00	10.10
A1	-	0.02	0.05	Ne	8.05BSC		
b	0.13	0.18	0.23	L	0.35	0.40	0.45
b1	0.12REF			E2	6.50	6.60	6.70
c	0.18	0.20	0.25	h	0.30	0.35	0.40
D	9.90	10.00	10.10	Nd	8.05BSC		
D2	6.50	6.60	6.70	Body Size	7.5*7.5		
e	0.35BSC						

Application Circuit